

THEOREM. *In an integral domain $\mathcal{A}$, for every polynomial P in the variable x with coefficients in $\mathcal{A}$, $P \in \mathcal{A}[x]$, and every $r \in \mathcal{A}$, there exist unique $Q \in \mathcal{A}[x]$ and $R \in \mathcal{A}$ such that*

$$P(x) = (x - r)Q(x) + R,$$

where

$$R = P(r).$$

○ Let $P_n(x)$ be a polynomial in $\mathcal{A}[x]$ of degree n ,

$$P_n(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where each coefficient $a_i \in \mathcal{A}$.

LEMMA. From any difference of two n -th powers, $x^n - c^n$ (where $n \geq 1$), $(x - c)$ can be factored out. That is, there exists a polynomial $Q_{n-1}(x)$ such that

$$x^n - c^n = (x - c)Q_{n-1}(x).$$

○ (by mathematical induction on n)

The **base case** $n = 1$ holds since

$$x^1 - c^1 = (x - c)(1),$$

where the constant $1 = Q_0(x)$ is a polynomial as well.

Inductive Hypothesis. Assume for some arbitrary positive integer $k \geq 1$ there exists a polynomial $Q_{k-1}(x)$ such that

$$x^k - c^k = (x - c)Q_{k-1}(x).$$

Induction Step. To show that the lemma holds for $k + 1$ (assuming it holds for k), consider

$$x^{k+1} - c^{k+1},$$

that can be rewritten by adding $0 = -xc^k + c^kx$ and grouping as

$$x^{k+1} - xc^k + c^kx - c^{k+1} = (x^{k+1} - xc^k) + (c^kx - c^{k+1}).$$

Factoring out the common components from each group gives

$$x(x^k - c^k) + c^k(x - c).$$

Applying the inductive hypothesis to substitute $(x - c)Q_{k-1}(x)$ for $(x^k - c^k)$ yields

$$x(x - c)Q_{k-1}(x) + c^k(x - c).$$

Factoring out the common multiplier $(x - c)$ results in

$$(x - c)(xQ_{k-1}(x) + c^k),$$

where $xQ_{k-1}(x) + c^k = Q_k(x)$ is a polynomial — since $Q_{k-1}(x)$ is a polynomial by the inductive hypothesis and polynomials are closed under multiplication by x and addition of constants.

This completes the induction step, proving the lemma for all $n \geq 1$. ●

Explicitly, the $Q(x)$ polynomials are

$$\begin{aligned}
 1 &= Q_0(x) \\
 x(1) + c^1 &= x + c &= Q_1(x) \\
 x(x + c) + c^2 &= x^2 + xc + c^2 &= Q_2(x) \\
 x(x^2 + xc + c^2) + c^3 &= x^3 + x^2c + xc^2 + c^3 &= Q_3(x) \\
 x(x^3 + x^2c + xc^2 + c^3) + c^4 &= x^4 + x^3c + x^2c^2 + xc^3 + c^4 &= Q_4(x) \\
 &\dots \\
 x(x^{n-2} + x^{n-3}c + \dots + xc^{n-3} + c^{n-2}) + c^{n-1} &= x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1} = Q_{n-1}(x) \\
 x(x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1}) + c^n &= x^n + x^{n-1}c + \dots + xc^{n-1} + c^n = Q_n(x)
 \end{aligned}$$

In more compact notation with the summation symbol

$$Q_n(x) = x^n + x^{n-1}c + \dots + xc^{n-1} + c^n = \sum_{i=0}^n x^{n-i}c^i$$

and

$$xQ_{n-1}(x) + c^n = x \sum_{i=0}^{n-1} x^{n-1-i}c^i + c^n = \sum_{i=0}^{n-1} x^{n-i}c^i + c^n = \sum_{i=0}^n x^{n-i}c^i = Q_n(x)$$

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